

## Research Article

# Understanding of Thermal Conductance of Thin Gas Layers

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Received 5 November 2012; Revised 12 December 2012; Accepted 13 December 2012

Academic Editor: Shuyu Sun

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We studied heat conduction in a thin gas layer at micro- and nanoscales between two straight walls by atomistic modeling. Since the Knudsen number is high while the gas may be not really rarefied, we use the generalized Enskog-Monte-Carlo method (GEMC) for simulations. The thermal conductivity of thin gas layer is reduced significantly with the decreased thickness of gas layer. We examined a few possible causes including the rarefied gas effect and the thermal inertia effect. Our careful simulations indicate that the temperature jump on wall surfaces and the properties changing significantly by the confined space are two dominating factors to the thermal conductivity reduction of thin gas layers.

## 1. Introduction

With the rapid developments of micro-/nanotechnologies for fabrication and manufacture, heat management plays a more and more important role in further developments of MEMS/NEMS. For an example of a micromachined Pirani sensor [1], its mechanism is to measure the gas pressure by sensing the thermal conductivity of gas near the microbeam/microplates. Therefore for improvements of accuracy of such devices, understanding of heat conduction characteristics at micro- and nanoscales becomes very important. In recent years, the storage capacity of hard disk drivers continues increasing; for instance the recording density has approached to 1 Tbit/in.<sup>2</sup>, which leads to a significant decrease of the flying height of the slider to approximately 3.5 nm [2–6]. Because of the high heat flux density produced by high friction between the flying slider and the rotating disk, the anomalous heat conduction of the gas thin layer in between restricts actual improvements of storage capacity of hard disk. Additionally heat managements and performance optimizations for electronic integrated circuit chips demands as well clear understandings of heat transport mechanism of gas layers at micro- and nanoscales [7–12].

Therefore heat transfer at micro- and nanoscales has been of great interests in the past ten years [1, 2, 5, 7–10, 12–15]. Heat transfer in thin gas layers differs from that at

large scale. The literature [14, 16] has demonstrated that the natural convection is negligible at micro scale using Monte-Carlo simulations, which means that in an enclosed space of microdevices, we need only consider the heat conduction. Sieradzki [17] discussed the effect of gas pressure on heat conduction, whose theoretical prediction agreed well in the continuum and the free molecular regimes, respectively, but was invalid in transition regime. Zhu and Ye [18] proposed a new slip model for slip flows extend to a high-Knudsen regime, developed an analytical approach for collisionless steady-state heat conduction inside a fully diffuse enclosure, discussed the effect of partially thermal accommodated walls on the heat conduction in transition regime, and verified them by DSMC simulations. Nevertheless, the new slip model in high-Knudsen regime can only be used to deduce an accurate average heat flux. Denpoh [19] studied the heat conduction in a gap between wafer and susceptor as 1D or 2D rarefied gas problems using DSMC, and considered effects of gas species, surface temperature, energy accommodation coefficient on heat conduction, whose results showed that the thermal conductivity declined with the decrease of  $1/Kn$ , consistent with the extended Smoluchowski equation [20]. However their results were for the rarefied gas, not for the micro- and nanoscale gas. The previous study have shown that the mechanism of gas flow and heat transfer at micro- and nanoscale might be different from that of

rarefied gas even though they are both in the same range of Kn [21]. Until now to the best knowledge of the authors, the dominating factor of size effect and the mechanism of thermal conductivity decline with the characteristic length of gas thin layers are still unclear.

In this work, we are to reveal the mechanism of the thermal conductance of thin gas layer using numerical modeling. The paper is organized as follows. In Section 2, we introduce a Monte Carlo method for real gas instead of rarefied gas, using which we will study the thermal conduction of gas thin layer. In Section 3, we will examine the possible factors that influence the effective thermal conductivity of gas thin layer at small scales to discuss and reveal the mechanisms. Finally we will draw conclusions in Section 4.

## 2. Numerical Method

Since the concerned gas in the thin layer may be not really rarefied, in this work, we adopt a Monte Carlo method based on the Enskog equation, the generalized Enskog Monte Carlo method (GEMC) [22], for dense gas developed by considering high density effect on collision rates and both repulsive and attractive molecular interactions for a Lennard-Jones fluid. The enhanced collision rate is determined by considering the excluded molecular volume and shadowing/screening effects based on the Enskog theory. The internal energy exchange model is also adapted to be consistent with the generalized collision model based on the Parker's formula. The equation of state for a nonideal gas is therefore derived involving the finite density effect and the van der Waals intermolecular force, changing from the Clapeyron equation to the van der Waals equation. More details about this algorithm can be found in [22].

GEMC will degrade to and be consistent with DSMC and other Monte Carlo methods for gas flows at really low densities. However for high densities, in contrast to the previous Monte Carlo approaches, the GEMC predictions agree better with experimental data for gas transport properties in a wide temperature region [22]. The GEMC method has been proved valid for both the ideal and nonideal gas flow and heat transfer [21]. Besides, since a generalized soft sphere model for collision is used in GEMC, the temperature ranges are greatly expanded to both low temperature regime and high temperature regime.

## 3. Results and Discussion

The simulation system is described as follows. Consider a nitrogen gas in a two-dimensional (2D) thin layer as shown in Figure 1. The pressure of the gas is at 1 atm so that the gas is not really rarefied. The temperature of lower and upper wall is given at  $T_1$  and  $T_2$ , respectively. If not specified,  $T_1$  and  $T_2$  are 500 K and 300 K, respectively, in this work. Periodic boundary conditions are implemented on the left and the right sides. The thickness of the thin layer is  $L$ , also called the characteristic length, ranging from 10 nm to 2000 nm in our simulations, which leads to a Kn variation from about 0.03 to 6.

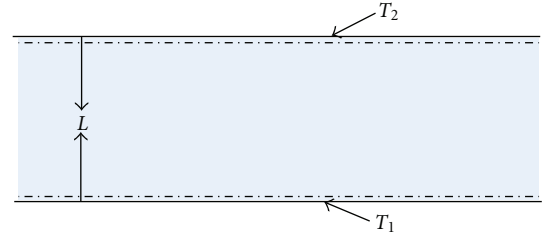


FIGURE 1: Sketch diagram of gas thin layer between two walls.  $L$  is the thickness of thin-layer, and  $T_1$  and  $T_2$  are the upper wall and the lower wall temperature, respectively.

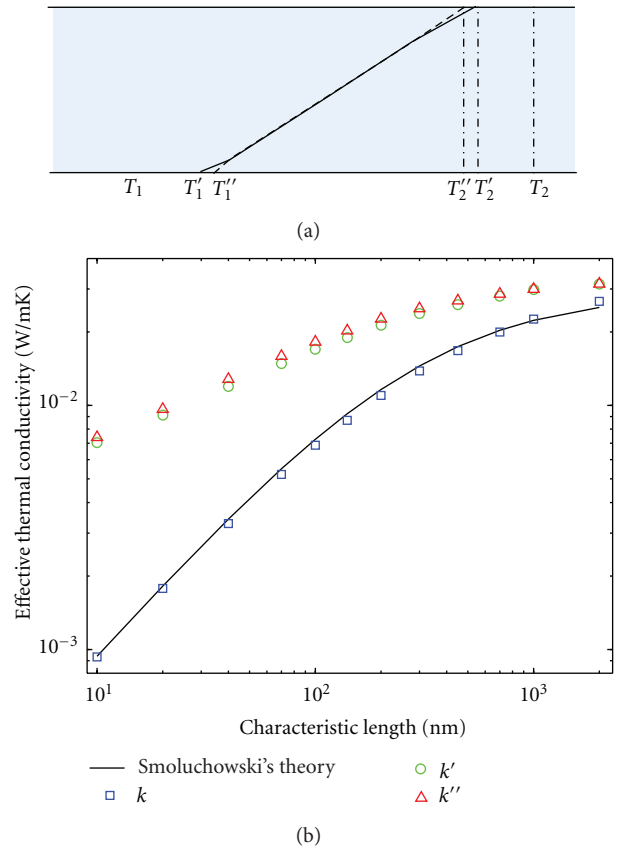


FIGURE 2: (a) Definitions of temperature, where  $T_1$  and  $T_2$  are the temperature of walls for calculation of  $k$ ,  $T'_1$  and  $T'_2$  are the temperature of gas adjacent to the walls for calculation of  $k'$ , and  $T''_1$  and  $T''_2$  are linearized wall temperature for calculation of  $k''$ ; (b) effective thermal conductivities as a function of thickness of gas layer calculated by different temperature gradients.

In the following parts of this section, we are to examine the possible reasons that decline the effective thermal conductivity of the thin gas layer with the thickness, and to discuss the mechanisms.

**3.1. Rarefied Gas Effects.** Based on the Fourier's Law, the thermal conductivity is calculated by  $k = -q/(\partial T/\partial x)$ . The

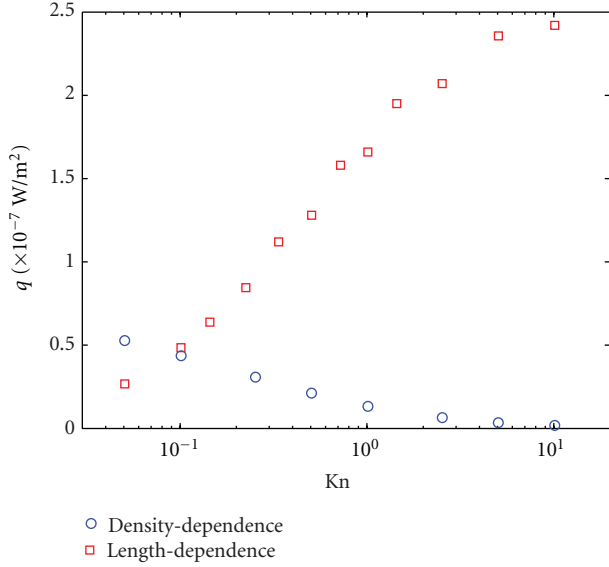


FIGURE 3: Heat flux across gas layers versus Kn for given walls temperature. The circles are from real rarefied gas effects, where the wall distance is given and the gas density varies. The squares are from size effects of micro scale gas, where the gas density is given (1 atm) and the thickness of gas layer varies. The variation trends are opposite.

heat flux  $q$  is calculated based on molecular collisions with the concerned surface [23–25]

$$q = \frac{[(\sum_{i=1}^n \epsilon_{tr} + \sum_{i=1}^n \epsilon_{rot})_i + (\sum_{i=1}^n \epsilon_{tr} + \sum_{i=1}^n \epsilon_{rot})_r] N_0}{\Delta t \cdot \Delta x}, \quad (1)$$

where the subscripts “ $i$ ” and “ $r$ ” denote the incident and reflected molecular streams, respectively,  $\epsilon_{tr}$  is the molecular translational energy,  $\epsilon_{rot}$  the rotational energy,  $N_0$  the number of gaseous molecules associated with a computational molecule,  $\Delta t$  the time period of sampling, and  $\Delta x$  the grid size of the surface. The temperature gradient is thus critical to the accuracy of the effective thermal conductivity. Usually the temperature gradient is calculated by  $\Delta T/L$  assuming a linear distribution of temperature across the system [26, 27]. However since the Kn number of the gas layer is high, the temperature jump on the wall surfaces is strongly suspected, which reduces the temperature difference in fact, to be a key factor of the decreased effective thermal conductivity.

The rarefied gas effect due to the high Kn leads to a nonlinear temperature distribution of gas across the layer, as shown in Figure 2(a). The temperature of gas adjacent to the walls,  $T_1'$  and  $T_2'$ , are different from the wall temperature,  $T_1$  and  $T_2$ , because of the temperature jump. The temperature of bulk gas, which is far away from the walls, seems still following a linear law. When fitting a linear relationship from the middle, we get two *linearized* wall temperature,  $T_1''$  and  $T_2''$ , by the intersection points between the fitting line and the wall surfaces. Thus we have three ways to calculate the effective thermal conductivity:  $k = qL/(T_1 - T_2)$ ,  $k' =$

$qL/(T_1' - T_2')$ , and  $k'' = qL/(T_1'' - T_2'')$  respectively. In the latter two formulas, the temperature jump does not contribute to the corresponding effective thermal conductivity.

Figure 2(b) depicts the effective thermal conductivities as a function of thickness of the gas layer calculated by the three different temperature differences. The symbols are from numerical simulations and the solid line is from the extended Smoluchowski equation for rarefied gas [20]. Our previous study showed that for nitrogen gas the gas density effect was negligible once the density is lower than 4.47 times of standard atmosphere pressure [21]. Therefore for the present cases (1 atm) the extended Smoluchowski model is still available as a benchmark. The results show that the effective thermal conductivity  $k$  calculated by the wall surface temperature difference ( $T_1$  and  $T_2$ ) agrees with the theoretical solution, and that the effective thermal conductivities,  $k'$  and  $k''$ , by the gas temperature difference ( $T_1'$  and  $T_2'$ ) and the *linearized* wall temperature difference ( $T_1''$  and  $T_2''$ ) are both higher than the theoretical prediction. The effect from the temperature jump is erased in the calculations of both  $k'$  and  $k''$ , however they still decrease with the thickness of gas layer as  $k$  does. It suggests that even though the gas in between the walls is at the same temperature and pressure as a free gas, the close walls change the bulk property indeed. Therefore the temperature jump on surface is an important factor for the decrease of the effective thermal conductivity of thin gas layer but may not be the only dominating reason of such a size effect.

As we mentioned above, even though the Knudsen numbers of the micro- and nanoscale gas and the rarefied gas are in the same range, the mechanisms of gas flow and heat transfer may be quite different [21]. Figure 3 shows the heat flux between two walls across the gas layer changing with the Knudsen numbers. We compare the results from two different causes in the same Kn range. For the real rarefied gas effect, we fix the wall distance and vary the gas density, while for the micro gas effect, the thickness of gas layer varies and the gas density is given as that at 1 atm. The results show opposite variation trends for such two cases, which proves again different mechanisms of thermal conductance of between rarefied gas and micro- and nanoscale gas, and also suggests that the rarefied gas effect is not the dominating factor of thermal conductance of thin gas layer.

**3.2. Effect of Thermal Inertia.** Besides the surface temperature jump, the temperature gradient increases significantly when the walls get closer for given walls temperature. If the temperature gradient is very high, the effective thermal conductivity may decline by the effect from thermal inertia based on a thermomass model [29]. Recent studies have shown that the thermal inertia effect causes the effective thermal conductivity of nanotubes and nanowires decreasing with size [30–32]. Therefore we are checking whether this effect also plays the key role in the thermal conduction in thin gas layer.

Figure 4 depicts the variations of effective thermal conductivity with the thickness of gas layers or with the temperature deference between walls. Figure 4(a) shows

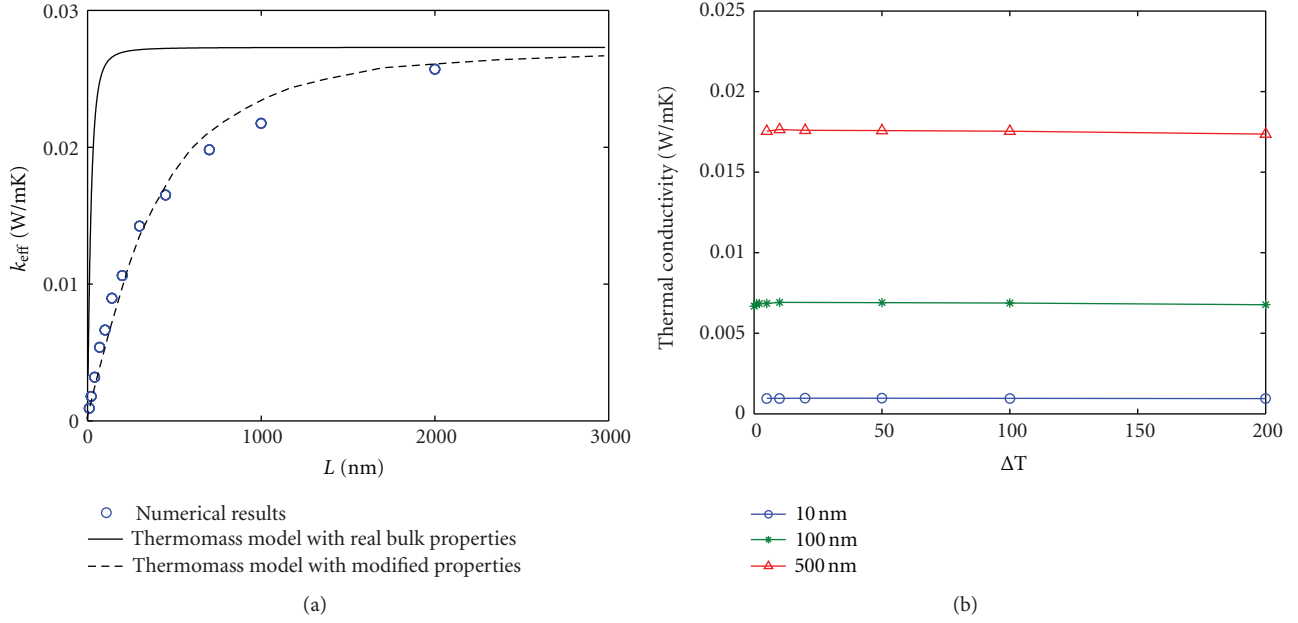


FIGURE 4: Comparisons with the thermomass model. (a) Effective thermal conductivity versus thickness of gas layer. The circles are the numerical data by GEMC and the lines are based on the thermomass model. The  $k_{\text{eff}}$  is calculated with  $T_1$  and  $T_2$ , and  $T_1 = 500$  K and  $T_2 = 300$  K. The real properties used in the thermomass model are:  $\rho = 0.85$  kg/m<sup>3</sup>,  $k = 0.0273$  W/m K,  $c_v = 747.5$  J/kg K and  $\gamma = 1.4$  [28]. The modified property used in the dashed line is  $c'_v = cv/7$ ; (b) effective thermal conductivity as a function of temperature difference for three different characteristic lengths.

the results of GEMC (symbols) compared with predictions from the thermomass model using the real properties or modified properties (lines). Even though both the numerical simulations and the thermomass predictions decline with the decreased thickness of gas layer, they deviate from each other significantly. However if we modified the specific heat capacity by a multiplier of  $1/7$ , the theoretical model agrees with the numerical data well as the dashed line shows. Up to now no theory or experimental data has clarified that the specific heat capacity of gas at small scale decreases to  $1/7$  of its value at normal scale. Therefore it is still a challenge to compare the thermomass model with the numerical data for thin gas layer yet.

Another important inference of the thermomass model is that the effective thermal conductivity of thin layer should decrease with increasing temperature gradient due to the enhanced thermal inertia effect. Therefore we checked such effects by our atomistic simulations. For three given thicknesses of thin gas layers, we changed the temperature difference while keep the average temperature as a constant at  $T = 400$  K. Figure 4(b) indicates that the effective thermal conductivity of gas layer is almost independent of the temperature gradient, which deviates again from the thermomass model. Therefore Figure 4 suggests that the mechanism of scale effect of thermal conductivity of thin gas layer does not relate to the thermal inertia, or the thermomass model has to be significantly developed further for such a case.

**3.3. Effects of Confined Space.** Although the thermal inertia seems nothing to do with the scale effect of thermal

conduction in thin gas layer, the discussions in 3.2 does give us some inspiration. The only difference between the gas in the thin layer and that in a bulk free space is the confined space by the close two walls. The confined space may change the gas properties to deviate from those in large space.

As a first step, the gas kinetic theory gives the thermal conductivity of gas by  $k = (1/D)c_v v l$  [33], where  $D$  is the effective dimension number,  $v$  the mean velocity and  $l$  the molecular mean free path. In the classical theory, the mean molecular velocity and specific heat capacity are constant for a given temperature, and the thermal conductivity is determined by the molecular mean free path. Therefore we examine the variation of mean free path first. The mean free path in a confined space is calculated as the average distance travelled by a moving particle between collisions with either another particle or the walls. When the walls get closer, the collision possibility between gas molecules and walls increases.

Figure 5(a) compares two molecular trajectories for different thickness of gas layer. For a layer thickness at 2000 nm, the molecule moves like a diffusive process and the walls influence its movement seldom. However when the layer thickness is close to the theoretical mean free path, the molecules collide with the wall frequently and the molecule travels as a ballistic way. Figure 5(b) shows that the molecular mean free path of gas in the thin layer decreases significantly with the reduced thickness of the gas layer. The mean free path of gas between walls is close to the value of gas in free space when the thickness of gas layer is larger than 1000 nm. The trend seems consistent with the thermal conductivity, which may suggest that the change of molecular mean free

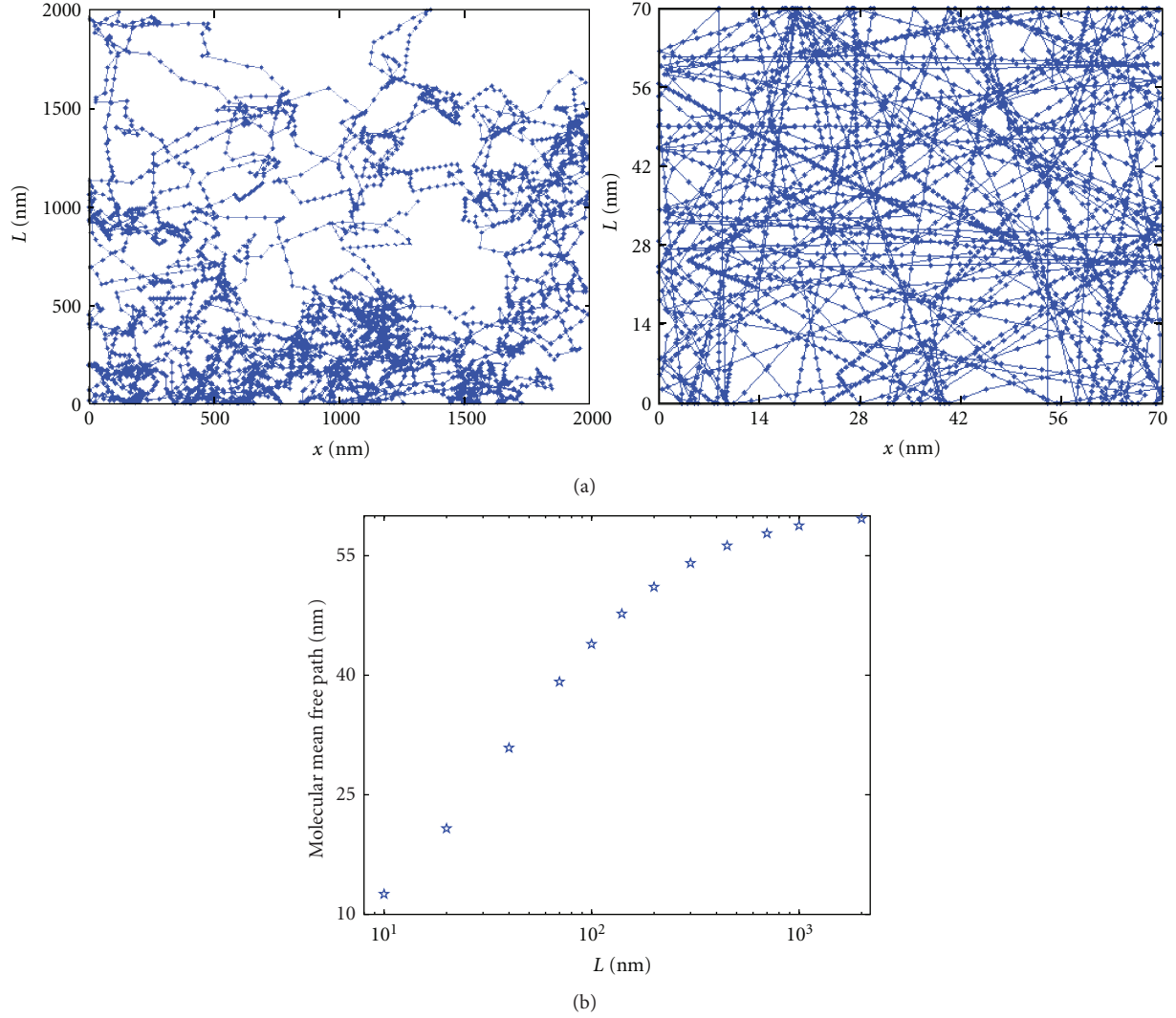


FIGURE 5: Size effect of gas molecular mean free path in the thin layer. (a) Molecular trajectory schematics for different thickness of gas layer. The left panel is for 2000 nm and the right panel is for 70 nm layer. The specular reflection is used on left and right sides for easy tracking and indication of trajectories, but the periodic boundary condition is implemented for calculations of other properties. (b) Mean free path versus thickness of gas layer. The wall temperatures are given at  $T_1 = 500$  K and  $T_2 = 300$  K.

path may be one of the key reasons for the scale effect of effective thermal conductivity of gas layer.

In conventional kinetic theories and thermodynamics, the specific heat capacity,  $c_v$ , is only dependent of temperature. No one has reported a scale effect on the specific heat capacity of gas in a thin layer yet. Based on the definition of  $c_v = (\partial E / \partial T)_v$  with  $E$  representing the internal energy, we could calculate  $c_v$  by

$$c_v = \frac{\langle \delta E^2 \rangle}{mk_b T^2} = \frac{1}{mk_b T^2} [\langle E^2 \rangle - \langle E \rangle^2], \quad (2)$$

where  $m$  is the molecular mass and  $k_b$  the Boltzmann constant [34, 35].

Figure 6(a) shows the calculated  $c_v$  changing with the thickness of the gas layer. Different from description of the conventional kinetic theory, the specific heat capacity is dramatically changed by the confined space between walls,

increasing significantly with the decreased thickness when the thickness of the gas layer. Even when the gas layer thickness is larger than 1000 nm, the calculated  $c_v$  is still much higher than the value at large scale; however the trend is to approach the normal value.

In gas kinetic theory [33], the mean velocity of gas molecules is only a function of temperature as  $v \propto \sqrt{RT}$ . When the gas molecules can only move in the confined space between two close walls, the mean velocity of gas molecules is also a function of the thickness of the wall distance, as shown in Figure 6(b). The mean velocity of gas molecules are calculated by averaging the particle velocities in the domain. The result indicates that a smaller thickness of gas layer leads to a lower mean velocity of gas molecules.

As a result, the effective thermal conductivity can be calculated based on the kinetic theory:  $k = c_v v l / D$  with  $D$

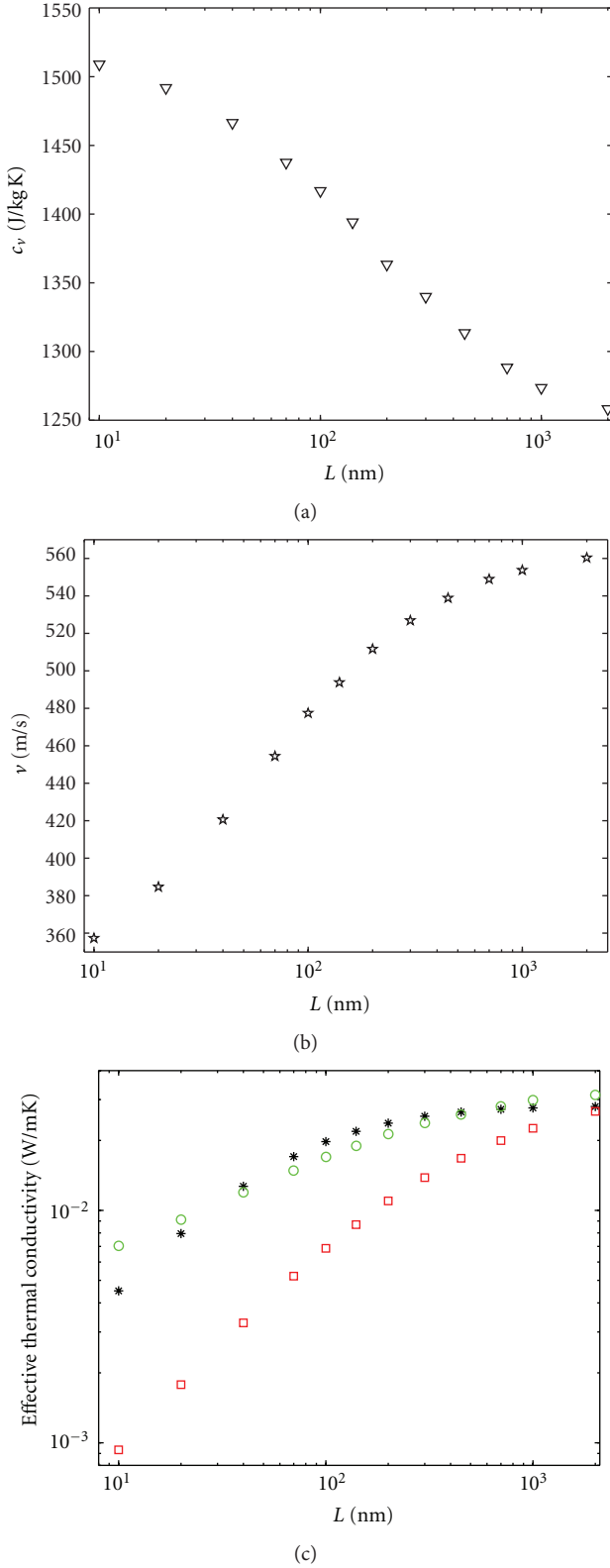


FIGURE 6: Effective properties of gas layer changed by the confined space. (a) Heat capacity versus the thickness of gas thin layer; (b) gas molecular mean velocity versus the thickness of gas thin layer; (c) effective thermal conductivity calculated by the kinetic theory (stars), compared with the modeling results using different temperature differences,  $k$  (squares) and  $k'$  (circles).

representing the effective dimension number. We use two-dimensional GEMC modeling in this study ( $D = 2$ ), while the studied cases are actually one-dimensional ( $D = 1$ ) for periodic boundaries on left and right sides. Therefore we adopt  $D = 1.5$  eclectically in Figure 6(c) for theoretical prediction (the stars). This method is phenomenological and not so rigorous, but surprisingly we find that the kinetic predictions with the adapted gas properties ( $c_v, v, l$ ) by the confined space agree quite well with the effective *bulk* thermal conductivity ( $k'$ ), where the surface temperature jump is excluded.

#### 4. Conclusions

The mechanism of thermal conductance of thin gas layer at micro/nanoscale has been numerically studied using the generalized Enskog-Monte-Carlo method in this work. The effective thermal conductivity decreases significantly with the reduced thickness of gas layer. When the distance between walls is as small as micro- and nanoscale, the Kn may be high and the temperature jump on surfaces is not negligible. Our simulations indicated that the effective thermal conductivity still declined significantly with the thickness of gas layer even if the temperature jump was excluded, which suggested that the temperature jump was not the only key factor for the size effect and the *bulk* thermal conductivity of gas in the layer was also changed. We then examined the effect from thermal inertia based on the thermomass model, but it seemed this model still needed further development for the concerned problem. Finally when considering the effects from the confined space on the gas effective properties, we found that the mean free path of gas showed the same variation trend as the effective thermal conductivity. The specific heat capacity and the mean molecular velocity, which were generally treated independent of size in classical kinetic theories and thermodynamics, also varied with the characteristic length in the confined micro- and nanoscale space. The results indicated that the gas effective properties, which were changed significantly by the confined space between walls, also dominated the thermal conductivity reduction of thin gas layers.

#### Acknowledgments

This work is financially supported by the Tsinghua University Initiative Scientific Research Program, the NSFC Grant (no. 51176089), and the startup funding for the Recruitment Program of Global Young Experts of China (no. 320503002).

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